Cisco PE Router Hardware/Environmental Fault Resolution SOP for Node ID: MG-PE-04

Node: MG-PE-04 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_5308

Description

The failure domain of Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) occurs on the PE role of Cisco equipment due to the inherent vulnerability of the ASIC memory to soft errors caused by cosmic radiation, alpha particles, or transient voltage noise. The likelihood of this failure is compounded by the high density and clock speeds of modern ASICs, which can amplify these errors. Furthermore, the PE role is critical in forwarding traffic between the customer edge and the provider edge, making the impact of this failure domain severe. In the case of MG-PE-04, running on Cisco equipment at the MG Road Core site, the Alarm Code ALARM_5308 signifies a Critical failure, necessitating immediate attention to prevent network disruptions.

Pre-requisite Checks

- Ensure remote hands are dispatched to the MG Road Core site to facilitate physical access to the equipment.

- Verify the status of redundant links and ensure they are operational.

- Gather all relevant logs and configuration files from the affected node.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to cosmic radiation or transient voltage noise	High
Line card crashes due to overheating or electrical instability	Medium
Power supply redundancy loss due to hardware failure or cabling issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support mg-PE-04, Expected Output: System information, configuration, and logs will be displayed., Description: This command will provide a comprehensive view of the system, including hardware and software configurations, and any relevant logs or error messages that might indicate the root cause of the failure.

Command: show debug mg-PE-04, Expected Output: Debugging information, including packet captures and system calls, will be displayed., Description: This command will provide detailed information about system calls, packet captures, and other low-level events that might help identify the source of the failure.
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Command: show environment mg-PE-04, Expected Output: Environmental information, including temperature and humidity readings, will be displayed., Description: This command will provide information about the environmental conditions around the equipment, which might be relevant to troubleshooting hardware or environmental faults.
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Resolution Steps

Step 1: Step: Replace the faulty line card or power supply, Command: card replace mg-PE-04 line-card slot 1, Description: If the failure is due to a faulty line card or power supply, replace the affected hardware with a spare unit, and ensure it is properly configured and cabled.

Step 2: Step: Reset the ASIC memory, Command: clear mg-PE-04 asic memory, Description: If the failure is due to ASIC memory parity errors, attempt to reset the memory to its default state, and verify that the issue is resolved.

Step 3: Step: Reboot the router, Command: reload mg-PE-04, Description: If none of the above steps resolve the issue, attempt to reboot the router in a controlled manner, and verify that the issue is resolved.

Post-Check Verification

```
Command: show mg-PE-04 status, Expected Output: The router will display a healthy status, with all interfaces and services online., Description: Verify that the router is functioning correctly, with all interfaces and services online, and that the issue is fully resolved.

Command: show mg-PE-04 logs, Expected Output: The logs will display no errors or warnings related to the original failure., Description: Verify that the logs display no errors or warnings related to the original failure, indicating that the issue is fully resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: configure replace mg-PE-04 startup-config, Description: If the fix fails, restore the original configuration from the backup file, and verify that the issue is resolved.

- Step: Revert the hardware changes, Command: card revert mg-PE-04 line-card slot 1, Description: If the fix fails, revert the hardware changes made during the resolution steps, and verify that the issue is resolved.